

Class 22: Subspace or Not?

Parallels between subspace proofs and linear transformation proofs

	Subspace H	Linear Transformation T
Zero Condition	H must contain $\vec{0}$	$T(\vec{0})$ must $= \vec{0}$
Addition Condition	If $\vec{u}$ and $\vec{v}$ are in H then $\vec{u} + \vec{v}$ must be in H .	$T(\vec{u} + \vec{v})$ must equal $T(\vec{u}) + T(\vec{v})$
Scalar multiple Condition	If $\vec{u}$ is in H and c is any scalar then $c\vec{u}$ must be in H .	$T(c\vec{u})$ must equal $cT(\vec{u})$

Practice

$$1. \quad W = \left\{ \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} : \begin{array}{l} a+b+c=d \\ 2a-b-d=0 \\ b+c=a \end{array} \right\} \quad \begin{array}{l} a+b+c-d=0 \\ 2a-b-d=0 \\ -a+b+c=0 \end{array}$$

W is the null space of the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 & -1 \\ 2 & -1 & 0 & -1 \\ -1 & 1 & 1 & 0 \end{bmatrix}$$

$$2. \quad H = \left\{ \begin{bmatrix} s+2t \\ 3s-t \\ 5s \end{bmatrix} : s, t \text{ are in } \mathbb{R} \right\}$$

H is the column space of

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \\ 5 & 0 \end{bmatrix}$$